Polynomials and FFT exercises

30-E01 Degree bounds and aliasing

If $\deg A \le 2$ and $\deg B \le 3$, how many distinct values determine their product? What is the smallest radix-2 length N? For $(1+x)^2$, explain why a length-2 forward transform, pointwise product, and inverse give `[2,2]` instead of `[1,2,1]`. Interpret the result modulo $x^2 - 1$.

30-E02 Every FFT intermediate value

Use the positive-exponent forward FFT on `[2,5,7,-1]`. List the even and odd two-point DFTs, the bit-reversed array, and all values after both stages. Give each butterfly's physical positions and twiddle. Distinguish original indices from physical positions in stage 1.

30-E03 Inverse sign and normalization

Prove the inverse formula from root orthogonality. For $a = [2, 5+i, 7, -1]$, what results from a forward FFT followed by another positive-exponent FFT and division by 4? Is this generally conjugation? What results if the negative sign is correct but division by 4 is omitted?

30-E04 Existence and uniqueness of interpolation

Prove existence of a degree-less-than-n interpolant at n distinct points using the Lagrange basis. Prove uniqueness using the root bound. Given $A(0) = 1, A(1) = 3$, exhibit infinitely many polynomials if only $\deg A < 3$ is promised. What changes if $\deg A < 2$?

30-E05 Three transforms for convolution

For $A = 1 + 2x, B = 1 - x, N = 4$, list every entry of DFT(A), DFT(B), their pointwise product, and the unnormalized inverse. Recover the coefficients. Count the butterflies and circuit stages for an $N = 8$ FFT. Explain the models behind serial work and parallel depth.

Polynomials and FFT exercises (continued)

30-E06 In-place overwrites and the stage invariant

A butterfly uses $u = 9, v = 4, w = 1$ to compute $u + wv$ and $u - wv$. Suppose the upper cell is overwritten first and the lower computation then reads that updated upper cell. What wrong value results? Give the correct temporary-variable version. Explain the blocks before and after every iterative FFT stage. Can this radix-2 program be applied directly to $n = 6$?

30-E07 NTT conditions and signed reconstruction

For $p = 17, N = 4, \omega = 13$, verify the root's order and the inverse of N. List the complete NTTs of `[1,2,0,0]` and `[1,-1,0,0]`, their pointwise product, and the inverse. Why can inverse output 15 mean -2? If only a coefficient bound of 20 is known, can the same modulo-17 output uniquely recover the integer coefficient?

> Complete the derivations before opening the separate solutions PDF.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study